

Knowledge Management Exam – 1h30 Duration

IKSEM Master – Paris Pantheon Sorbonne University – May 20th, 2025 -

Exercise 1: (5 points)

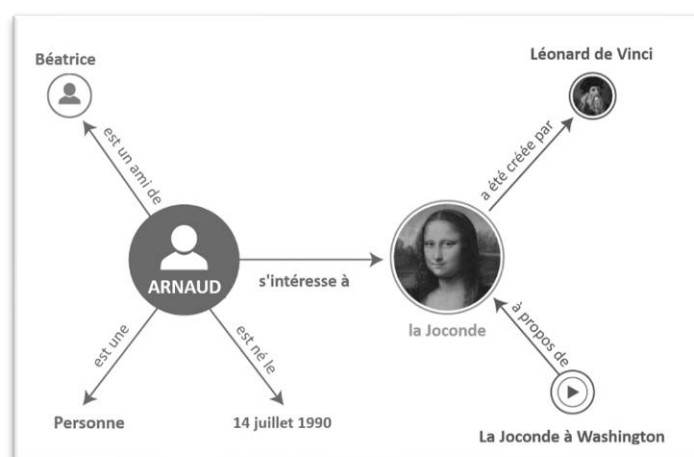
Question 1. Draw the RDF graph corresponding to the following description given in this RDF/XML:

```
<?xml version="1.0"?>
<!DOCTYPE rdf:RDF [
  <!ENTITY xsd "http://www.w3.org/2001/XMLSchema#" >
  <!ENTITY rdfs "http://www.w3.org/2000/01/rdf-schema#" >
  <!ENTITY rdf "http://www.w3.org/1999/02/22-rdf-syntax-ns#" >
]>

<rdf:RDF
  xmlns="http://www.polymtl.ca#"
  xml:base="http://www.polymtl.ca#"
  xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema#">
  <A rdf:ID="x1">
    <p rdf:parseType="Collection">
      <rdf:Description rdf:ID="x2"/>
      <rdf:Description>
        <q rdf:resource="x3"/>
      </rdf:Description>
    </p>
  </A>
</rdf:RDF>
```

Question 2. List all the triples in the following RDF Graph

Question 3. Propose an RDF/XML representation of this RDF Graph



Exercise 2: (3 points)

Illustrate the DIKW pyramid through a practical example in the context of improving guest satisfaction at a hotel.

Exercise 3: (4 points) Case Study: Knowledge Management in Hotel Operations

A luxury hotel chain, Elite Stays Hotels, operates multiple properties worldwide. To maintain high service standards, improve guest satisfaction, and streamline operations, the hotel implements a robust Knowledge Management Framework.

Propose examples (recommendations and tools) at each stage of the 5 components of the KM Framework to help the company reaching its objectives.

Exercise 4. MCQ (8 points)

1. What is the main goal of the Semantic Web?
A) To improve internet speed
B) To make web data machine-readable and interoperable
C) To replace traditional web pages
D) To increase social media engagement

2. Which language is commonly used to represent knowledge in the Semantic Web?
A) HTML
B) XML
C) RDF
D) SQL

3. What does OWL stand for in the context of the Semantic Web?
A) Online Web Language
B) Open Web Logic
C) Ontology Web Language
D) Object Web Linking

4. What is the function of SPARQL in the Semantic Web?
A) It is a web design language
B) It is a query language for retrieving and manipulating RDF data
C) It is a protocol for secure web transactions
D) It is used for web page styling

5. Which of the following is a key principle of the Semantic Web?
A) Using relational databases for all data storage
B) Replacing human-readable text with machine code
C) Using structured metadata to describe resources
D) Eliminating hyperlinks from web pages

6. What role do ontologies play in the Semantic Web?
A) They define rules for website layouts
B) They establish a common vocabulary for data integration
C) They improve website security
D) They act as web search engines

7. Which of the following are triple components in RDF?
A) Subject
B) Predicate
C) Object
D) Context

8. Which technologies are foundational to the Semantic Web?

- A) RDF
 - B) OWL
 - C) SPARQL
 - D) HTTP
-

9. What are the benefits of using ontologies in the Semantic Web?

- A) Enable data interoperability
 - B) Improve web page loading speed
 - C) Provide a shared understanding of concepts
 - D) Ensure data security
-

10. Which of the following statements about RDF are true?

- A) RDF represents data as triples
 - B) RDF is based on XML
 - C) RDF is used for styling web pages
 - D) RDF stores data in relational tables
-

11. What are common use cases of the Semantic Web?

- A) Data integration across different platforms
 - B) Natural language processing
 - C) Website design improvements
 - D) Decentralized knowledge representation
-

12. What does a SPARQL query return?

- A) A list of RDF triples
 - B) A web page layout
 - C) JSON, XML, or CSV-formatted data
 - D) An image file
-